

# DUAL-MOTOR PMSM DYNAMOMETER PLATFORM

## PLATFORM, ELECTRICAL & INTERFACES

TI LAUNCHXL-F28069M, dual BOOSTXL-DRV8305EVM stages, encoders, CAN, power, and physical verification

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 2.4 | 5 AUGUST 2026

### DOCUMENT PURPOSE

Defines the controlled platform, electrical, interface, safety, verification, and acceptance requirements. Embedded algorithms, host software, CAN message handling, custom ADC code, and efficiency results are controlled by the companion Firmware & Host FRD.

# Functional Requirements Document Framework

## WHY THIS IS AN FRD

This document defines the physical and electrical behavior the platform shall provide, the interfaces and constraints under which it shall operate, and the objective evidence required for acceptance. Vendor diagrams and photographs support verification; they do not transfer ownership of third-party board designs.

## Purpose, Scope, and System Boundary

This FRD provides a customer-facing public description of the dynamometer platform, its physical system boundary, major electrical elements, evidence basis, and qualification limits. Detailed implementation and deployment records are intentionally withheld.

# 1. Platform Architecture

The platform combines a TI C2000 controller, two three-phase inverter stages, two mechanically coupled encoder-equipped PMSMs, a shared bench supply, and a real-time measurement workflow. The public description is limited to the system architecture and integration boundary.

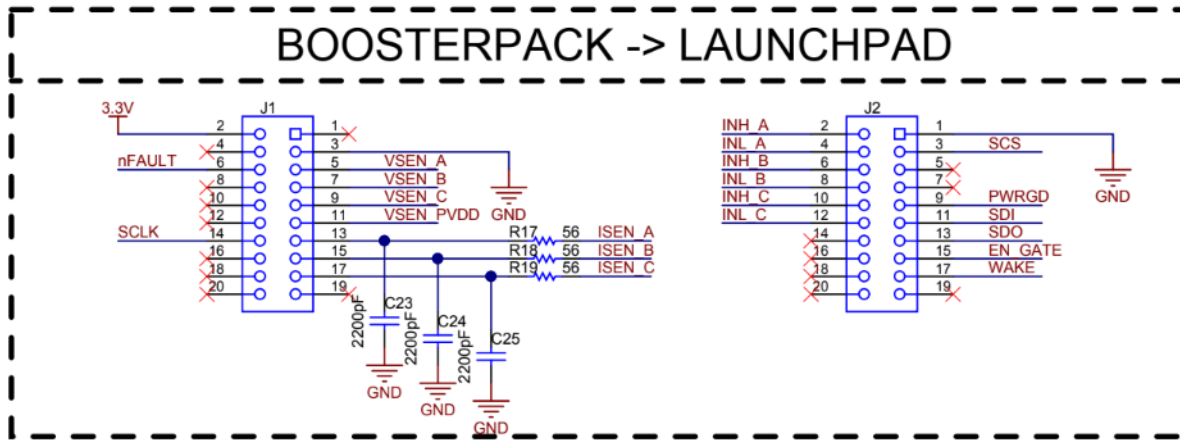


Figure 1. BoosterPack-to-LaunchPad board-interface arrangement.

## 1.1 Primary Elements

| Element               | Quantity | Public role                                                       |
|-----------------------|----------|-------------------------------------------------------------------|
| LAUNCHXL-F28069M      | 1        | Embedded controller, encoder interfaces, PWM/ADC and CAN endpoint |
| BOOSTXL-DRV8305EVM    | 2        | Three-phase inverter stages for drive and load machines           |
| Encoder-equipped PMSM | 2        | Mechanically coupled drive and load machines                      |
| 24 V DC supply        | 1        | Current-limited bench power source                                |
| CAN interface         | 1        | Supervisory physical-layer connection                             |

## Integrated setup views

These photographs document the physical integration boundary used for the public hardware description: the controller, dual inverter stages, cabling, and interface hardware are shown as an assembled test setup rather than as isolated board references. Original source photographs are retained in [media/physical\\_setup/original/](#).

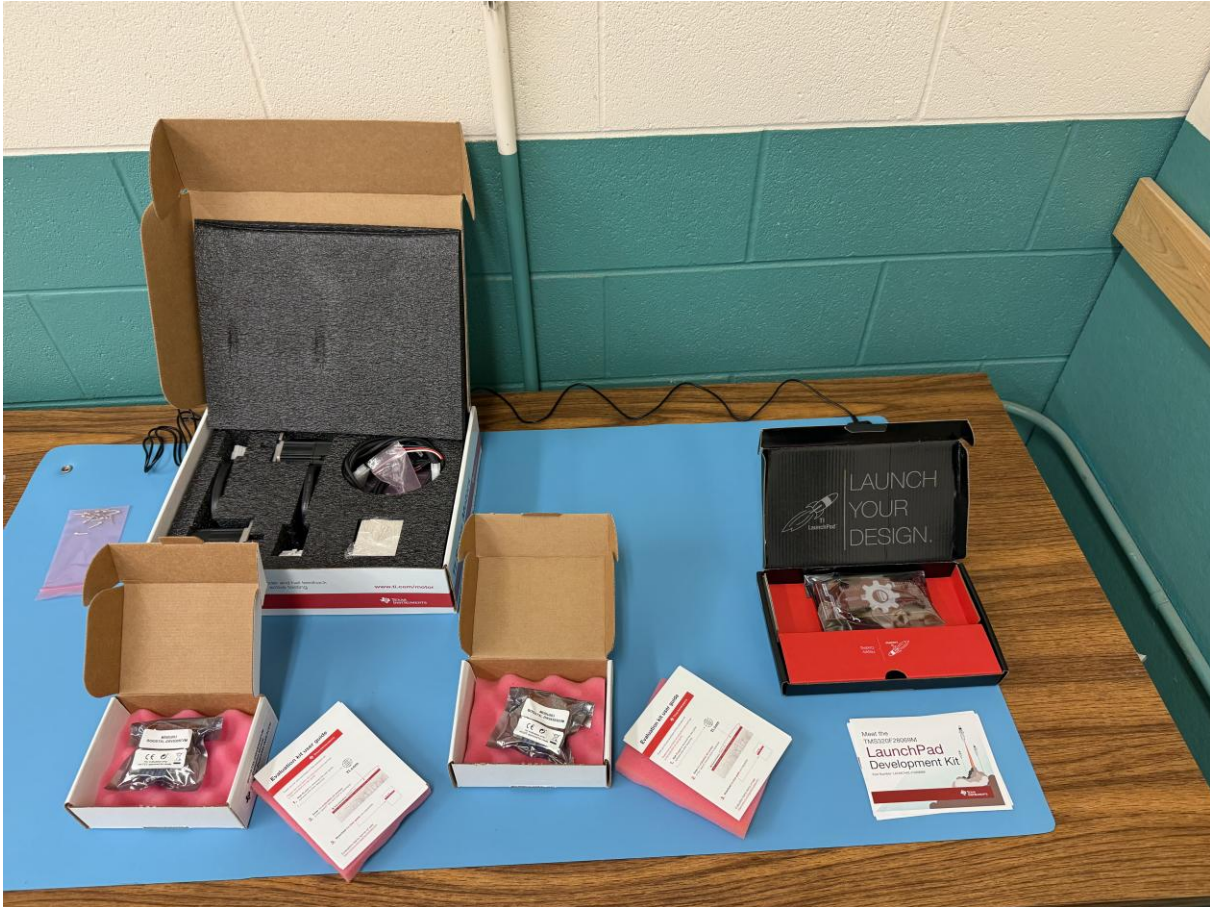


Figure 2. Integrated setup components: controller, inverter, cable, and interface hardware arranged for system integration.



Figure 3. TI C2000 LaunchPad and connected interface cabling.

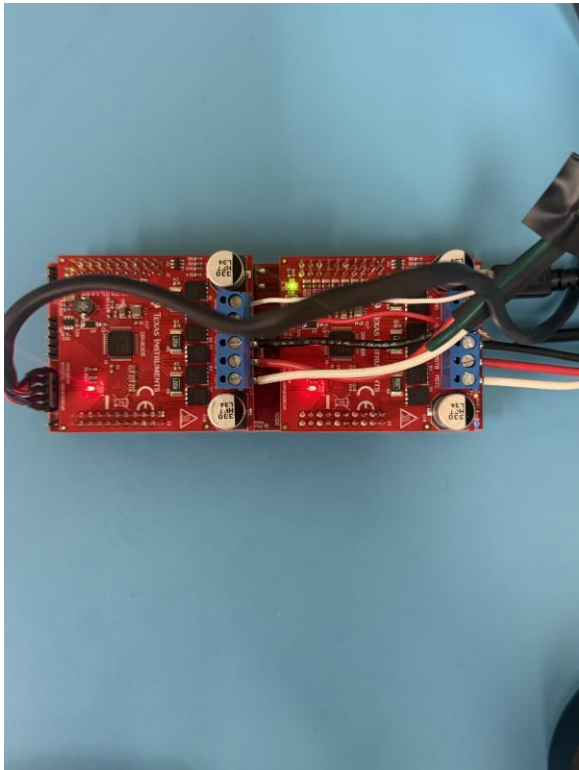


Figure 4. Two inverter boards shown with their connected cabling.

2. Platform and Electrical Requirements

| ID      | The system shall...                                                                                                                                         | Verification | Status       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| PLT-001 | The platform shall provide two mechanically coupled encoder-<br>Dual-Motor PMSM Dynamometer - Platform and Electrical FRD • Controlled engineering document | Inspection   | Demonstrated |

| ID      | The system shall...                                                                                             | Verification     | Status       |
|---------|-----------------------------------------------------------------------------------------------------------------|------------------|--------------|
|         | equipped PMSMs.                                                                                                 |                  |              |
| PLT-002 | The controller stack shall support two independent three-phase inverter stages.                                 | Inspection       | Demonstrated |
| PLT-003 | The platform shall retain physical access for deployment, diagnostics, and emergency shutdown.                  | Inspection       | Demonstrated |
| PLT-004 | Rotating components shall be restrained and kept clear of hands during operation.                               | Procedure review | Required     |
| PLT-005 | The public release shall distinguish institution-owned or third-party hardware from authored models and source. | Document review  | Verified     |

## 2.1 Power and Motor Interface Requirements

## 3. Interfaces and deployment boundary

### 3.1 Public release boundary

## 4. Schematics, BOM, and Ownership Boundary

Vendor and third-party material is acknowledged only at a high level. Detailed schematics, bills of material, connector maps, and manufacturing information are not included in this public FRD.

## 5. Verification and Acceptance

Platform verification was considered across physical inspection, interface review, controlled measurement, feedback observation, and communication checks. Detailed procedures and configuration records are withheld from this public FRD.

### 5.1 Open Qualification Items

- Independent torque and electrical-power calibration are outside this release.
- Formal EMC, vibration, ingress, environmental, and long-duration thermal qualification are not included.
- A complete independent protection chain and certified stop architecture are not claimed.
- No vendor PCB manufacturing sources, Gerbers, pick-and-place files, or fabrication packages are redistributed.

### 5.2 Release Gate

## Inherited platform and station boundary

The platform includes provided or inherited hardware and third-party support material. This release documents integration work and does not claim authorship of vendor board designs or private laboratory records.

Station-specific procedures, records, identifiers, and deployment details are outside this public release.